Homework #2

This is the homework for week 3. The due date is 11:59 p.m., September 22. Late submission will automatically result in "0" grade. Please include codes for the program part in your submission.

1 Computation Questions

(**Question1):** (20 Points) Suppose $X_1, X_2, \cdots, X_{50}$ are i.i.d. Poisson distributed random variables with parameter $\lambda = 3$. Approximate the probability of $\sum_{i=1}^{50} X_i \in [1, 10]$ using central limit theorem.

For the mean and the variance of Poisson distribution are both λ, the central limit theorem says

$$\frac{1}{\sqrt{n\lambda}} \sum_{i=1}^{n} (X_i - \lambda) \to_d N(0, 1).$$

In particular,

$$\sum_{i=1}^{50} X_i \in [1, 10] = \sum_{i=1}^{50} (X_i - 3) \in [-149, -140] = \frac{1}{\sqrt{150}} \sum_{i=1}^{50} (X_i - 3) \in [-12.16, -11.43] \approx 0.$$

(**Question2):** (20 Points) Let Y be $\chi^2(n)$ distributed random variable. Derive the limiting distribution of

$$W = (Y - n)/\sqrt{2n} \text{ as } n \to \infty.$$

Suppose Z_i are i.i.d. normal random variables, we know that

$$\sum_{i=1}^{n} Z_i^2 \sim \chi^2(n).$$

Moreover,

$$\mathbf{E}Z_i^2 = 1 \text{ and } Var(Z_i^2) = \mathbf{E}Z_i^4 - 1 = 2.$$

Therefore, from Central limit theorem,

$$W = \frac{1}{\sqrt{n \times 2}} \sum_{i=1}^{n} (Z_i^2 - 1) \to_d N(0, 1).$$

(**Question3):** (40 Points) Suppose $Z_1, Z_2, \cdots, Z_n$ are i.i.d. Poisson distributed random variables with parameter λ.

i. Suppose A and B are independent Poisson distributed random variables with parameter λ_1 and λ_2. Prove that

$$A + B \text{ satisfies Poisson distribution with parameter } \lambda_1 + \lambda_2.$$

ii. Use induction to show that $\sum_{i=1}^{n} Z_i$ satisfies Poisson distribution with parameter $n\lambda$.

iii. Derive the asymptotic distribution of

$$\sqrt{n}(\frac{1}{n}\sum_{i=1}^{n} Z_i - \lambda).$$

i. Since for any $k \geq 0$,

$$Prob(A+B \leq k) = \sum_{q=0}^{k} Prob(A = q \cap B = k-q) = \sum_{q=0}^{k} \frac{\lambda_1^q}{q!}\exp(-\lambda_1) \times \frac{\lambda_2^{k-q}}{(k-q)!}\exp(-\lambda_2)$$

$$= \frac{1}{k!}\exp(-\lambda_1-\lambda_2)\sum_{q=0}^{k}\binom{k}{q}\lambda_1^q\lambda_2^{k-q} = \frac{(\lambda_1+\lambda_2)^k}{k!}\exp(-\lambda_1-\lambda_2),$$

and we prove the result.

ii. For $n = 1$, the result holds true. Suppose the result holds for $n = 1, 2, \cdots, n_0$, for $n = n_0 + 1$,

$\sum_{i=1}^{n_0+1} Z_i = (\sum_{i=1}^{n_0} Z_i) + Z_{n_0+1}$ satisfies Poisson distribution with parameter $n_0\lambda + \lambda = (n_0+1)\lambda$.

iii. For $\mathbf{E}Z_i = \lambda$,

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(Z_i - \lambda) \to_d N(0, \lambda).$$

2 Programming Questions

(20 points)

Implement the following experiment using Jupyter or R:

i. For the sample size $n = 5, 10, 20, 50$, generate $Z_i, i = 1, 2, \cdots, n$ Poisson distributed random variables with parameter 1. Calculate $A = \frac{1}{n}\sum_{i=1}^{n} Z_i$ and $B = \sqrt{n}(A-1)$.

ii. Repeat step 1 for 200 times, plot histograms of A and B. Overall, you should demonstrate 8 histograms, 4 for A and 4 for B.